Finding Time Complexity of Algorithms

Problem 1: Finding Complexity using Counter Method

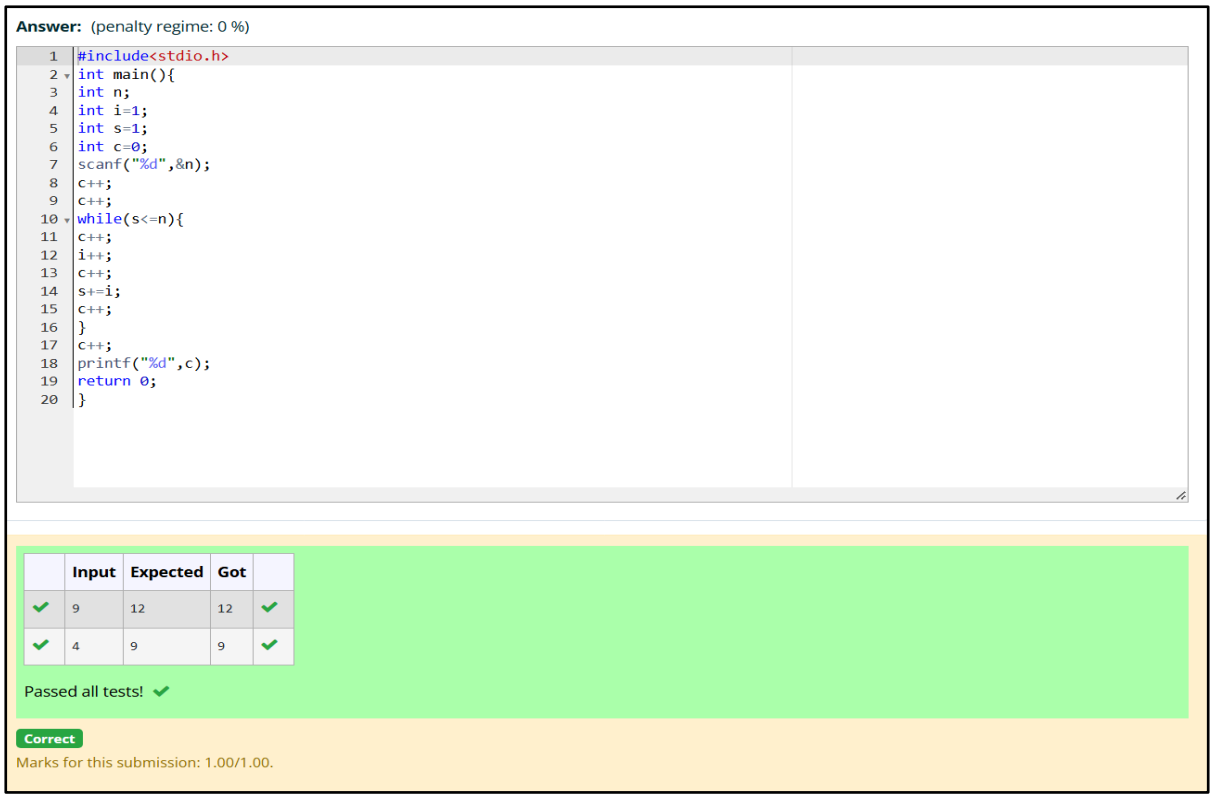
Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n){
   int i= 1;
   int s =1;
   while(s <= n){
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n
Output:
Print the value of the counter variable

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Problem 2: Finding Complexity using Counter method

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n){
if(n==1){
    printf("*");
  }
  else{
   for(int i=1; i<=n; i++){
     for(int j=1; j<=n; j++){
       printf("*");
       printf("*");
       break;
     }
   }
```

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}

}

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input: A positive Integer n

Output: Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    int c=0;
    scanf("%d",&n);
    c++;
    if(n==1){
        printf("*");
    }
    else{
        for(int i=1;i<=n;i++){
            c++;
            c++;
            for(int j=1;j<=n;j++){
                c++;
                c++;
                c++;
                break;
            }
        }
        c++;
    }
    printf("%d",c);
    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Problem 3: Finding Complexity using Counter Method

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i){
    if (num % i== 0){
        printf("%d ", i);
      }
    }
```

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}

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
int main(){
    int n;
    int c=0;
    scanf("%d",&n);
    for(int i=1;i<=n;i++){
        c++;
        c++;
        if(n%i==0){
            c++;
        }
    }
    c++;
    printf("%d",c);
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Problem 4: Finding Complexity using Counter Method

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n){
   int c= 0;
   for(int i=n/2; i<n; i++)
      for(int j=1; j<n; j = 2 * j)
         for(int k=1; k<n; k = k * 2)
            c++;
}
```

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Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n
Output:
Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
int n;
scanf("%d",&n);
int c=0;
c++;
for(int i=n/2;i<n;i++){
    c++;
for(int j=1;j<n;j=2*j){
    c++;
for(int k=1;k<n;k=k*2){
    c++;
c++;
}
c++;
}
c++;
}
c++;
printf("%d",c);
return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Problem 5: Finding Complexity using counter method

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n){
  int rev = 0, remainder;
  while (n != 0) {
      remainder = n % 10;
      rev = rev * 10 + remainder;
      n/= 10;
   }
print(rev);
```

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}

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n
Output:
Print the value of the counter variable

Answer:

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int c=0;
    int r=0,rem;
    c++;
    while(n!=0){
        c++;
        rem=n%10;
        c++;
        r=r*10+rem;
        c++;
        n/=10;
        c++;
    }
    c++;
    c++;
    printf("%d",c);
    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Greedy Algorithms

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